

费马引理

点 x_0 在某个邻域 $U = N_\delta(x_0)$ 内有定义, 且在 x_0 可导, 如果对任意 $x \in U$, 有

$$f(x) \leq f(x_0) \quad (\text{或 } f(x) \geq f(x_0))$$

$$\text{则 } f'(x_0) = 0$$

罗尔定理

(i) 在 $[a, b]$ 上连续

(ii) 在 (a, b) 上可导

(iii) $f(a) = f(b)$

则在 (a, b) 内至少有一点 ξ 使得 $f'(\xi) = 0$

例: 证明 $4ax^3 + 3bx^2 + 2cx = a + b + c \quad \exists x \in (0, 1)$ 使等式成立.

$$f(x) = ax^4 + bx^3 + cx^2 - (a+b+c)x.$$

根据罗尔定理.

$$f(0) = f(1) = 0$$

故存在 ξ s.t. $f'(\xi) = 0, \xi \in (0, 1)$.

拉格朗日中值定理

(i) 闭区间 $[a, b]$ 上连续

(ii) 开区间 (a, b) 上可导

定义辅助函数 $F(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} (x - a) + f(a) \right)$

1) 在 $[a, b]$ 上连续 2) 在 (a, b) 上可导 3) $F(a) = F(b) = 0$

由罗尔定理 $\exists \xi \in (a, b)$ s.t. $f'(\xi) = 0$

$$f'(\xi) = \frac{f(b) - f(a)}{b - a} = 0$$

$$f(b) - f(a) = f'(\xi)(b - a) \quad \xi \in (a, b)$$



$$f(x + \Delta x) = f(x) + f'(x + \theta \Delta x) \cdot \Delta x \quad (0 < \theta < 1) \quad \text{有限增量公式}$$

ξ, θ 与 x 的关系复杂, 尽量不求导

单侧导数极限定理 (充分不必要)

设 $\Delta > 0$, $f(x)$ 在 $N_{\Delta}(x_0)$ 内可导, 在点 x_0 连续. 如果 $\lim_{x \rightarrow x_0^+} f'(x)$ 存在或为无穷大, 则 $f'_+(x_0) = \lim_{x \rightarrow x_0^+} f'(x)$

$$\text{证明 } \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0^+} f'(x_0 + \theta(x - x_0)) = \lim_{x \rightarrow x_0^+} f'(x) \quad (0 < \theta < 1)$$

$$\text{则 } f'_+(x_0) \text{ 存在 且 } f'_+(x_0) = \lim_{x \rightarrow x_0^+} f'(x)$$

$$\text{例 } f(x) = \begin{cases} x^3 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$\begin{aligned} x \neq 0 \quad f'(x) &= 3x^2 \sin \frac{1}{x} + x^3 \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right) \\ &= 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x} \end{aligned}$$

$$\lim_{x \rightarrow 0^+} f'(x) = 0 \Rightarrow f'_+(0) = 0 \Rightarrow f'(0) = 0$$

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} \\ 0 \end{cases} \quad \lim_{x \rightarrow 0^+} \text{ 不存在 但 } \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{x^2 \sin \frac{1}{x}}{x} = 0 \quad \text{可导}$$

例. 可导函数导函数不可能有第一类间断点

$$\lim_{x \rightarrow x_0^-} f'(x) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \quad \textcircled{1} \quad f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0} \quad \textcircled{2} \quad f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}$$

在 x_0 处连续. 且 $\lim_{x \rightarrow x_0} f'(x)$ 存在 $f'(x_0) = \lim_{x \rightarrow x_0} f'(x) \Rightarrow \textcircled{1} = \textcircled{2}$

拉格朗日中值定理与不等式

$$0 < a < b \quad \text{证: } 1 - \frac{a}{b} < \ln b - \ln a < \frac{b}{a} - 1$$

$$\ln b - \ln a = \frac{1}{\xi} (b - a), \quad (a < \xi < b)$$

柯西中值定理

$[a, b]$ 上连续

(a, b) 内可导

在区间 (a, b) 上 $g'(x)$ 处处不为零

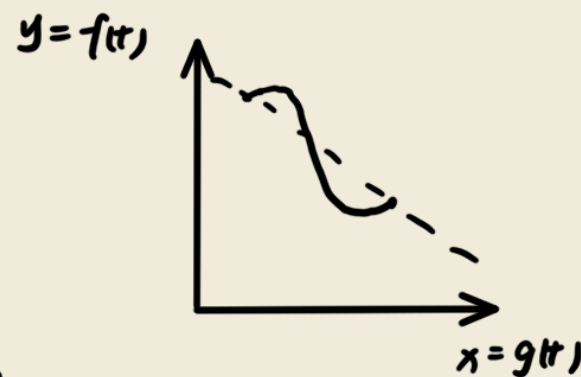
法- $\begin{cases} x = g(t) \\ y = f(t) \end{cases}$ 对 $y = f \circ g^{-1}(x)$ 用拉格朗日

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

法= $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$

$$F(x) = f(x) - \left(\frac{f(b) - f(a)}{g(b) - g(a)} (g(x) - g(a)) \right)$$

$$F'(\xi) = 0 = f'(\xi) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(\xi)$$



洛必达法则 $\frac{0}{0}$

(i) $x \rightarrow a$ 时 $f(x) \rightarrow 0, g(x) \rightarrow 0$

(ii) 在点 a 的去心邻域内可导, 即 $f'(x), g'(x)$ 存在, 且 $g'(x) \neq 0$

(iii) $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = k$ 有限或 ∞

证明 $f(a) = g(a) = 0$

$f(x), g(x)$ 在 $(a-\delta, a+\delta)$ 上连续

$f(x), g(x)$ 在 $(a-\delta, a), (a, a+\delta)$ 上可导

$$\frac{f(x)}{g(x)} = \frac{\overset{f(x)}{f(x)-0}}{\underset{g(a)}{g(x)-0}} = \frac{f'(\xi)}{g'(\xi)} \quad (\xi \text{ 介于 } a \text{ 与 } x \text{ 之间})$$

$$x \rightarrow a \Rightarrow \xi \rightarrow a, \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(\xi)}{g'(\xi)}$$

局限性. $\lim_{x \rightarrow 0} \frac{x^2 \cos \frac{1}{x}}{x} = \lim_{x \rightarrow 0} x \cos \frac{1}{x} = 0$

洛: $\frac{2x \cos \frac{1}{x} - \sin \frac{1}{x}}{1}$ 极限不存在.

$$\frac{\infty}{\infty} \text{ 取 } t = \frac{1}{x} \rightarrow 0 \quad \lim_{t \rightarrow 0} \frac{f(\frac{1}{t})}{g(\frac{1}{t})} = \lim_{t \rightarrow 0} \frac{f'(\frac{1}{t})(-\frac{1}{t^2})}{g'(\frac{1}{t})(-\frac{1}{t^2})} = \lim_{x \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

思路: $\frac{f(x)}{g(x)} = \frac{f(x)-f(b)}{g(x)-g(b)} \cdot \frac{(g(x)-g(b))}{g(x)} + \frac{f(b)}{g(x)}$

柯西 $\frac{f'(\xi)}{g'(\xi)} \quad \xi \text{ 介于 } x \text{ 与 } b \text{ 之间}$

先令 $x \rightarrow a \quad \frac{g(b)}{g(x)} \rightarrow 0 \quad \frac{f(b)}{g(x)} \rightarrow 0$

再令 $b \rightarrow a$.

严格证明 考虑 $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = k = 0$.

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = 0 \Rightarrow \forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } \forall b, 0 < |b-a| < \delta, \forall x \text{ 介于 } a, b \text{ 之间}$$

$$\left| \frac{f(x)}{g(x)} \right| \leq \left| 1 - \frac{g(b)}{g(x)} \right| \cdot \left| \frac{f'(x)}{g'(x)} \right| + \frac{|f(b)|}{|g(x)|}$$

$$\text{由 } \lim_{x \rightarrow a} \frac{f(b)}{g(x)} = 0 \text{ 知 } \exists \delta_2 < \delta \text{ s.t. } \left| \frac{f(b)}{g(x)} \right| < \frac{\varepsilon}{3}, \forall 0 < |x-a| < \delta_2$$

$$\text{由 } \lim_{x \rightarrow a} \frac{g(b)}{g(x)} = 0 \text{ 知 } \exists \delta_3 < \delta \text{ s.t. } \left| \frac{g(b)}{g(x)} \right| < \frac{\varepsilon}{3}, \forall 0 < |x-a| < \delta_3$$

$$\left| \frac{f(x)}{g(x)} \right| \leq \left(1 + \frac{\varepsilon}{3}\right) \cdot \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon, \quad \forall |x-a| < \min\{\delta_2, \delta_3\}$$

对于一般的 k

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} - k \right) = \lim_{x \rightarrow a} \frac{f(x) - k g(x)}{g(x)} = \lim_{x \rightarrow a} \left(\frac{f'(x)}{g'(x)} - k \right) = 0$$

$$\infty_1 - \infty_2 = \frac{1}{\frac{1}{\infty_1}} - \frac{1}{\frac{1}{\infty_2}} = \frac{\frac{1}{\infty_2} - \frac{1}{\infty_1}}{\frac{1}{\infty_1} \cdot \frac{1}{\infty_2}}$$

泰勒公式 $P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + \dots + a_n(x-x_0)^n$

当 $x \rightarrow x_0$ 时 $f(x) - P_n(x) = o((x-x_0)^n)$

$$a_0 = f(x_0) \quad a_1 = \frac{f'(x_0)}{1!} \quad \dots \quad a_n = \frac{f^{(n)}(x_0)}{n!}$$

$$R_n(x) = f(x) - P_n(x)$$

$$\text{余项 } R_n(x_0) = R'_n(x_0) = \dots = R_n^{(n)}(x_0) = 0$$

$$P_n(x) = \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x_0) (x-x_0)^k$$

条件 $f^{(n)}(x_0)$ 存在

$$\lim_{x \rightarrow x_0} \frac{R_n(x)}{(x-x_0)^n} = \lim_{x \rightarrow x_0} \frac{R'_n(x)}{n(x-x_0)^{n-1}} = \dots = \lim_{x \rightarrow x_0} \frac{R_n^{(n-1)}(x) - R_n^{(n-1)}(x_0)}{n! (x-x_0)} = \frac{1}{n!} R_n^{(n)}(x_0) = 0$$

导数的定义



↑
(n-1) 次洛必达

Taylor I 若 $f^{(n)}(x_0)$ 存在, 则

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + o((x-x_0)^n)$$

↑
佩亚诺余项

Taylor II 在含有 x_0 的邻域 U 内有直到 $(n+1)$ 阶的导数

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k + R_n(x)$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} \quad (\xi \in (x, x_0))$$

↑
拉格朗日余项

$$\text{当 } n=0 \text{ 时 } f(x) = f(x_0) + f'(\xi)(x-x_0)$$

$$\text{证: } R_n(x) = f(x) - \sum_{k=0}^n \frac{1}{k!} f^{(k)}(x_0) (x-x_0)^k$$

$$\frac{R_n(x) - R_n(x_0)}{(x-x_0)^{n+1} - (x_0-x_0)^{n+1}} = \frac{R'_n(\xi_1)}{(n+1)(\xi_1-x_0)^n} = \dots = \frac{R_n^{(n)}(\xi_n)}{(n+1)!(\xi_n-x_0)} = \frac{R_n^{(n)}(\xi_n) - R_n^{(n)}(x_0)}{(n+1)!(x_0-x_0) - 0} = \frac{R_n^{(n+1)}(\xi)}{(n+1)!} = \frac{f^{(n+1)}(\xi)}{(n+1)!}$$

↑
 $n+1$ 次柯西定理

n 阶麦克劳林公式

$$(1) e^x = 1 + x + \frac{1}{2!}x^2 + \dots + \frac{1}{n!}x^n + \frac{e^{\theta x}}{(n+1)!}x^{n+1}$$

$$(2) \sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots + (-1)^{m-1} \frac{1}{(2m-1)!}x^{2m-1} + \frac{\sin(\theta x + \frac{2m+1}{2}\pi)}{(2m+1)!}x^{2m+1} \quad (\text{展至 } 2m \text{ 次})$$

$$(3) \cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots + (-1)^m \frac{1}{(2m)!}x^{2m} + \frac{\cos(\theta x + (m+1)\pi)}{(2(m+1))!}x^{2(m+1)}$$

$$(4) \frac{1}{1-x} = 1 + x + x^2 + \dots = \sum_{n=0}^{\infty} x^n$$

$$(5) \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n$$

$$(6) (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots = \sum_{n=0}^{\infty} C_n^\alpha x^n$$

$$(7) \tan x = x + \frac{1}{3}x^3 + \frac{2}{15}x^5 + \frac{17}{315}x^7 + \dots$$

$$(8) \arcsin x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \dots = \sum_{n=0}^{\infty} \frac{(2n)!}{(2^n n!)^2 (2n+1)} x^{2n+1}$$

$$(9) \arccos x = \frac{\pi}{2} - \arcsin x = \frac{\pi}{2} - x - \frac{1}{6}x^3 - \frac{3}{40}x^5 - \dots = \frac{\pi}{2} - \sum_{n=0}^{\infty} \frac{(2n)!}{(2^n n!)^2 (2n+1)} x^{2n+1}$$

$$(10) \arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$\sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$\cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

设 $m > 1$. 求 $\lim_{x \rightarrow \infty} ((x^m + x^{m-1})^{\frac{1}{m}} - (x^m - x^{m-1})^{\frac{1}{m}})$

$$= \lim_{x \rightarrow \infty} x \left[\left(1 + \frac{1}{x}\right)^{\frac{1}{m}} - \left(1 - \frac{1}{x}\right)^{\frac{1}{m}} \right]$$

$$= \lim_{t \rightarrow 0} \frac{(1+t)^{\frac{1}{m}} - (1-t)^{\frac{1}{m}}}{t}$$

$$= \lim_{t \rightarrow 0} \frac{(1 + \frac{1}{m}t + o(t)) - (1 - \frac{1}{m}t + o(t))}{t}$$

$$= \frac{2}{m}$$

